

MSDM5004 Spring 2026

Homework 2 Part II

Due Mar. 15

3. Write a code using MATLAB (or other programming language) to compute the derivative of $f(x) = \frac{e^x}{x}$ at $x = 2$ using the central difference formula, and check the order of convergence.

4. Write codes using MATLAB (or other programming language) to solve the ODE

$$y' = 2t^{-2}(ty - 2y^2), \quad t \in [2, 4],$$

with initial condition $y(2) = 2$, using the following two numerical schemes: (1) forward Euler method and (2) backward Euler method. Choose time step $h = 1/128$.

5. Consider the equation

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}.$$

Show that the numerical scheme

$$\frac{U_j^{n+1} - U_j^{n-1}}{2\Delta t} = \frac{2(U_{j+1}^n - 2U_j^n + U_{j-1}^n)}{(\Delta x)^2}$$

is second order both in time and space.